

Name: vrajesh

*# Network Threats and Mitigation & Physical Security
and Risk*

*#### 1. Create a table listing 5 common network threats
(such as phishing, DDoS, man-in-the-middle, malware,
and password attacks), and for each, write one real-
world example from an app or service you use (like
WhatsApp, Instagram, or Paytm).*

Answer:

<i> Network Threat</i>	<i> Description</i>
<i> Real-World Example</i>	
<i> </i>	

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| *Phishing* | *Fake emails, messages, or websites that trick users into revealing passwords or banking details. | A fake **Paytm** SMS asking users to click a link and enter their login credentials.*

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| *DDoS (Distributed Denial of Service)* | *Attackers flood a server with huge amounts of traffic to make it unavailable. | An **Instagram** server experiencing heavy fake traffic, causing users to be unable to log in temporarily. |*

| *Man-in-the-Middle (MITM)* | *An attacker secretly intercepts communication between two users or devices. | Using **WhatsApp** on an unsafe public Wi-Fi where an attacker tries to intercept network traffic. |*

| *Malware* | *Harmful software that steals data, damages files, or controls a device.*

| *Downloading a fake **WhatsApp APK** that installs malware on a phone.* |

| *Password Attack* | *Attackers try many passwords to gain access to an account.*

| *Someone repeatedly tries different passwords to log into an **Instagram** account.* |

Explanation:

These threats target user accounts, personal information, and online services. Companies use encryption, firewalls, authentication, and monitoring systems to protect users from these attacks.

2. Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN

code by checking all possible combinations against a preset value.

*****Constraint:**** The script should print the number of attempts it took to guess the correct PIN.*

Answer:

Python Code

```python

Brute Force PIN Simulation

correct_pin = "4829"

attempts = 0

```
for i in range(10000):  
    guess = str(i).zfill(4)  
    attempts += 1
```

```
    if guess == correct_pin:  
        print("PIN Found:", guess)  
        print("Attempts:", attempts)  
        break  
    ...
```

Sample Output

```
    ...  
  
    PIN Found: 4829  
  
    Attempts: 4830  
    ...
```

Explanation:

- * The program checks every possible PIN from **0000** to **9999**.*
- * It compares each PIN with the correct PIN.*
- * When the correct PIN is found, it prints the PIN and the total number of attempts.*

3. List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.

Answer:

1. Unauthorized Access

Risk: An unauthorized person may enter the server room and access important servers or networking devices.

Mitigation: Install biometric authentication or keycard access so that only authorized staff can enter.

2. Fire

Risk: A fire can damage servers, routers, switches, and important company data.

Mitigation: Install smoke detectors, fire alarms, and an automatic fire suppression system.

3. Power Failure

Risk: A sudden power outage can shut down servers and may cause data loss or service interruption.

Mitigation: Use a UPS (Uninterruptible Power Supply) and a backup generator to provide continuous power.

4. Theft of Equipment

Risk: Servers, routers, switches, or storage devices may be stolen from the server room.

Mitigation: Install CCTV cameras, lock the server racks, and keep the server room secured at all times.

Explanation:

Physical security protects servers and networking equipment from damage, theft, or unauthorized access.

Proper security measures help ensure that online services continue to operate safely.

4. Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).

Answer:

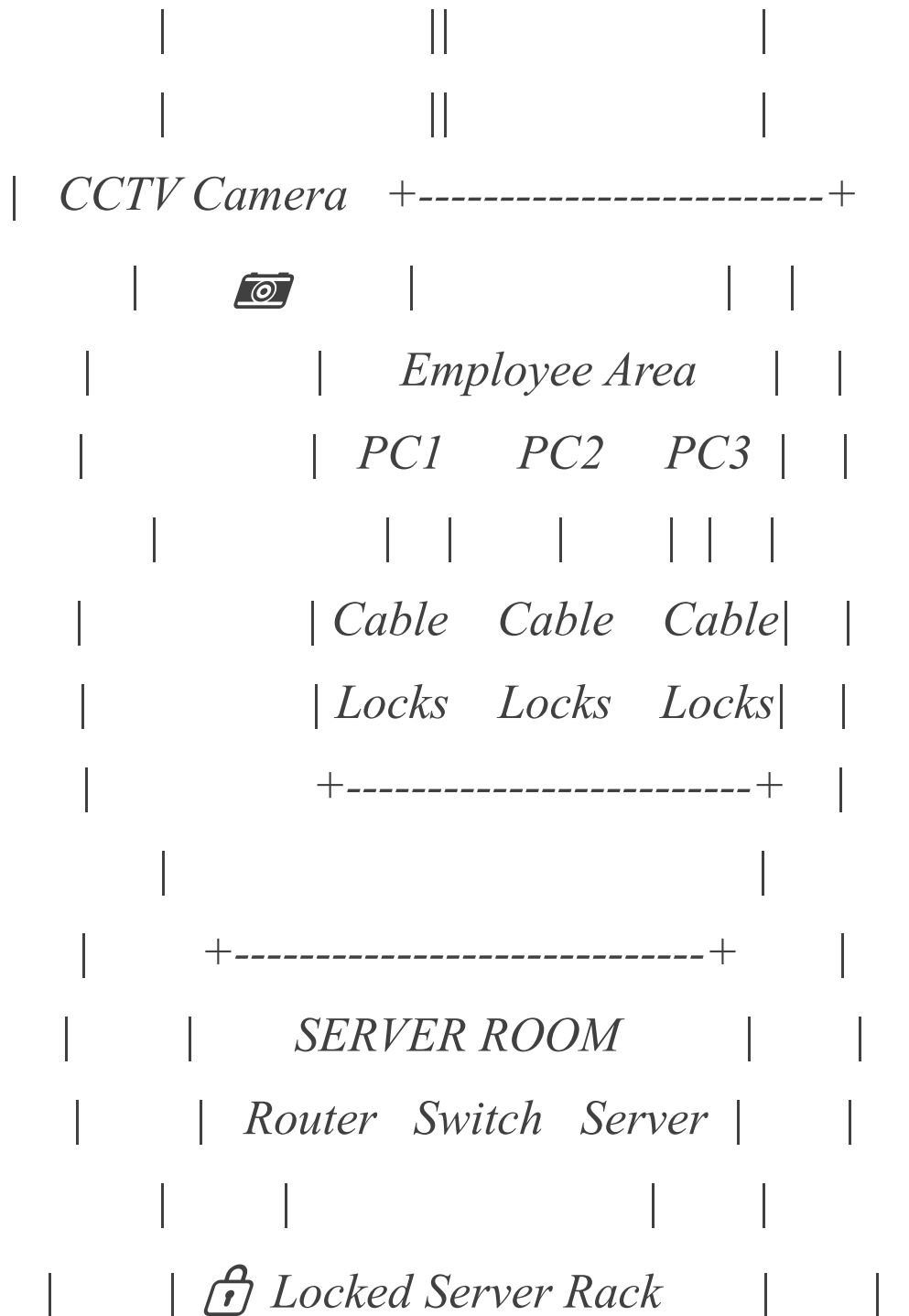
Office Floor Plan

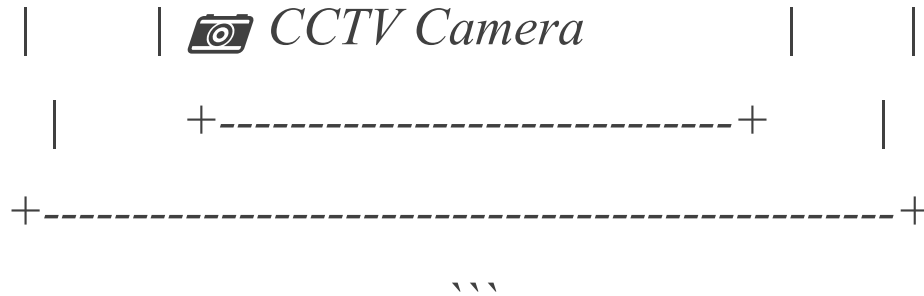
``text

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| *OFFICE ENTRANCE* |

| *[Biometric Access Door]* |





Physical Security Measures Used

1. ***Biometric access** at the entrance to allow only authorized people.*
2. ***CCTV cameras** to monitor the office and server room.*
3. ***Cable locks and locked server racks** to prevent theft of computers and networking equipment.*

5. Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.

Answer:

College Computer Lab Security Checklist

- 1. Install and regularly update antivirus software on every computer.*
- 2. Enable Windows Firewall on all lab computers.*
- 3. Use strong passwords and change administrator passwords regularly.*
- 4. Lock the server room with biometric or keycard access.*

5. *Install CCTV cameras to monitor the computer lab and server room.*
6. *Regularly update the operating system and all software to fix security vulnerabilities.*
7. *Back up important student records and lab data to a secure location on a regular schedule.*

Explanation:

*This checklist combines both ****network security**** and ****physical security**** practices. Following these steps helps protect the college computer lab from cyber attacks, unauthorized access, hardware theft, and data loss.*